

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

9 Further Human–Computer Interaction (lec40)

You are designing a new syntax for a programming language like Java, with the intention of making it more approachable to students by using English words instead of punctuation symbols.

Lecture 1: using theory in the design process

- (a) How does an HCI designer use a theory of human behaviour in their design practice? [2 marks]

Answer: HCI designers are interested in making interaction ‘better’. Theories provide a way of criticising designs to guide inventing new ones.

Alternative uses of theory, for example, in supporting communication between designers are also acceptable.

Lecture 7: evaluating interactive systems

- (b) What is an empirical method? Give an example and short description of that method. [3 marks]

Answer: An empirical method is a method based on observation of a phenomena in the world.

Example: Think aloud study Description: Participants in a study are asked to use a system, and to describe what they are thinking as they do so. Their descriptions are analysed to describe their mental model of the system.

Full marks for the description of any relevant empirical method

Lecture 8: designing complex systems

- (c) Describe in terms of the Cognitive Dimensions two trade-offs involved between the existing Java syntax and the new proposed syntax. [8 marks]

Answer: The change from punctuation to words will increase role expressiveness, making it easier to understand the meaning of the program, but will also increase difuseness. This will improve reading/searching at the expense of exploration.

The increase in difuseness will also increase viscosity by making changes require longer edits to the characters. This will make exploration slower.

For each trade-off: 1 mark for the activity 2 marks for naming dimensions that the tradeoffs are between 1 marks for accurate characterisation of the tradeoffs

Full marks for any reasonable tradeoff described within this or a similar structure.

Lecture 8: designing complex systems

- (d) How does the programming environment relate to this analysis? [2 marks]

Answer: The interface determines the possible interactions with the notation. For example, a development with syntax highlighting will make it easier for developers to read large amounts of code, and a code-completion feature would make edits to the code quicker decreasing the increased viscosity of the code.

Full marks for a discussion of the effect of the interface on each tradeoff

Lecture 7: evaluating interactive systems

- (e) A manager makes a proposal to try different permutations of syntaxes and

measure students' performance using each permutation. How would you measure and compare the students' performance? [2 marks]

Answer: Measure the amount of time it takes students to correctly complete a development task using each syntax.

Comparison would be done using a statistical significance test such as the sign-test, between the different syntax options.

Full marks for any reasonable measurement and comparison method

Lecture 7:
evaluating
interactive
systems

(f) What are the likely strengths and weaknesses of the approach in Part (e)? [3 marks]

Answer: This method will provide an internally valid study method for determining which syntax is more efficient to use. However it is likely to be dominated by the learning effects and natural variance meaning that a large number of participants would be needed. It also doesn't provide a description of why the results appeared the way they did, so the method doesn't guide future improvements to the design.

Full marks for any reasonable critical analysis of the approach
